

→ Greedy, DP, Divide and Conquer, Brute force,

Backtracking

N-queen problem

N=1



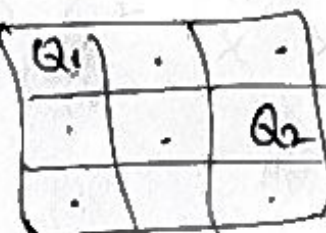
N=2



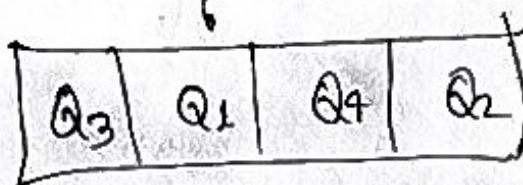
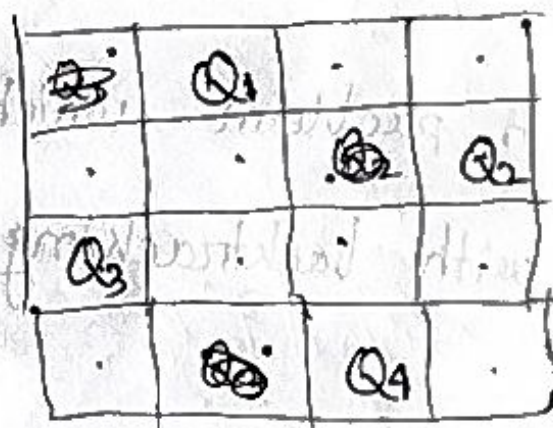
X

we cannot put 2 Queen

N=3



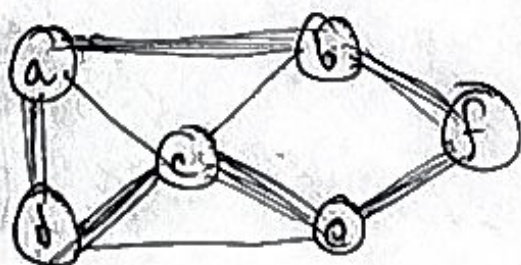
N=4



single dimension

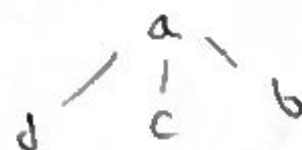
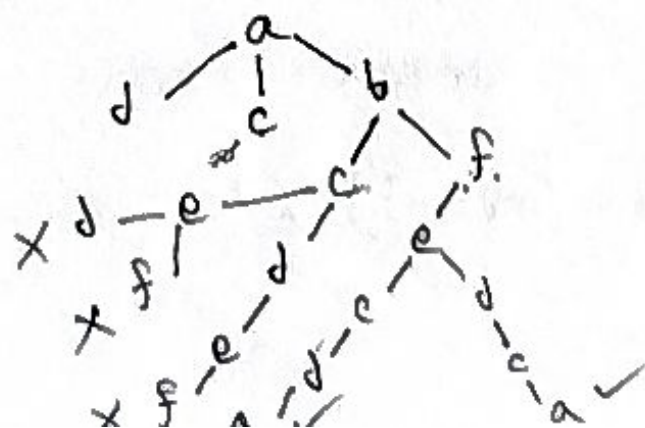
one solution

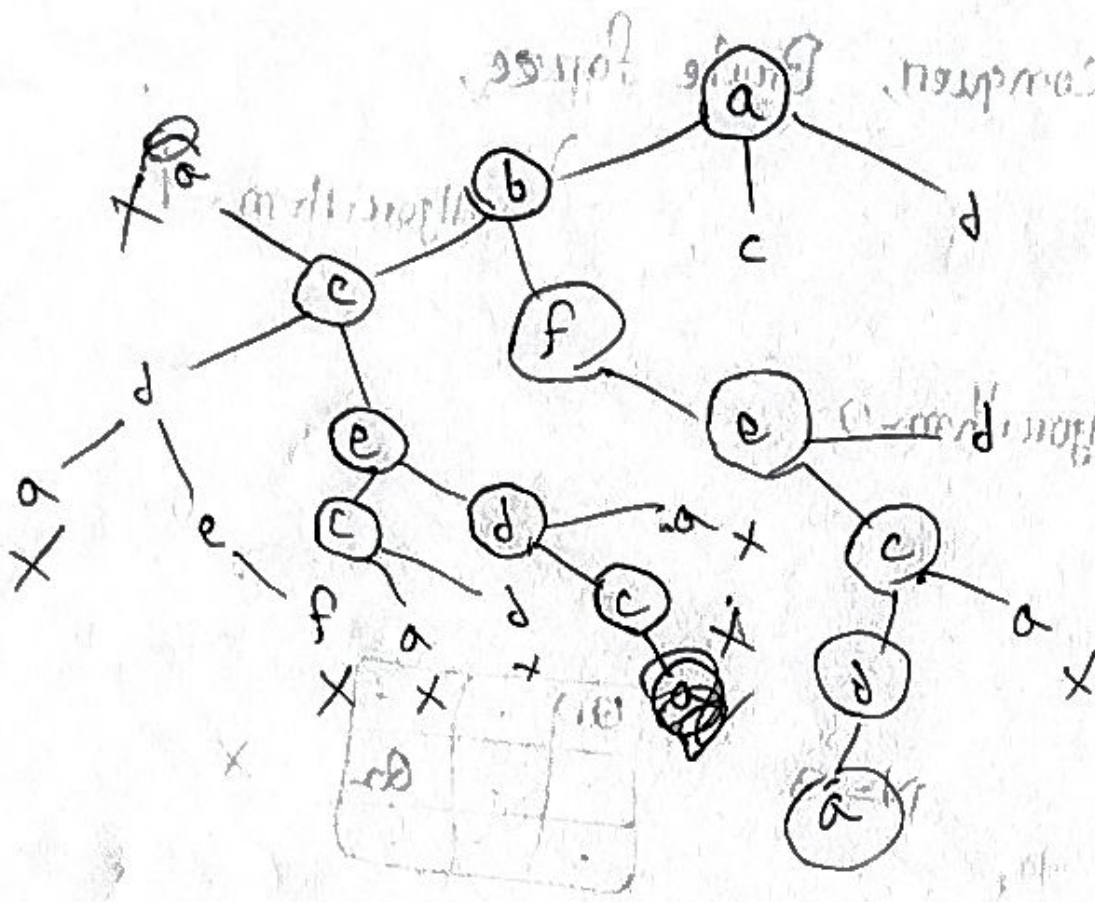
Hamiltonian Circuit



a b f e c d a

alternate path
explore to
possible places



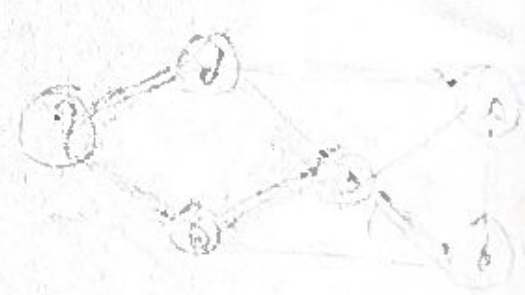


Amotz Levitin
3rd edition
The Design and Analysis
of Algorithms

A problem which can be solve in polynomial time
with backtracking.

while (rem $\neq$ 0)
rem = a % b

a = b
b = rem



Algorithm - (N-Queen)

NQueen(k, n):

for $i = 1$ to n do:

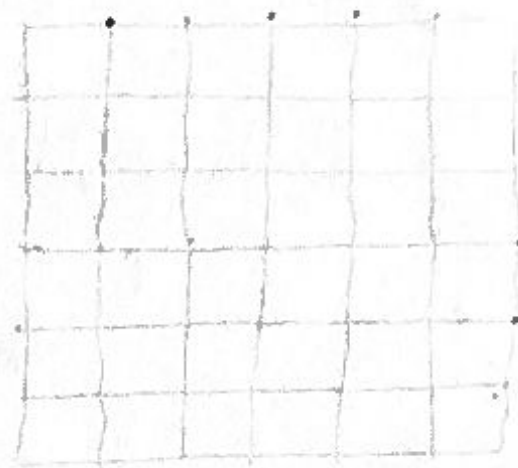
if (place(k, i)) then:

$x[k] = i$

if ($k == n$) then print:

$x[1 \dots n]$

else NQueen($k+1, n$)



1	2	3	4	5
---	---	---	---	---

--	--	--	--	--

place(k, i):

for $j = 1$ to $(k-1)$:

if ($x[j] == i$ or $(\text{abs}(x[j] - i) == \text{abs}(j - k))$)

return false

return true

Linear time Algorithm for NQueen

i) approach is not backtracking

ii) pattern based linear time algorithm

iii) $i = 1$ to $N/2$ even number column

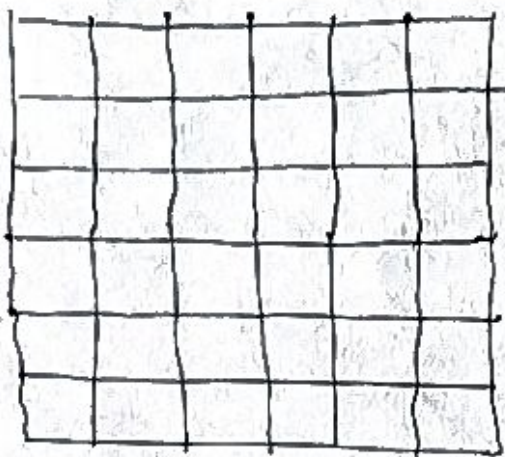
iv) $i = N/2 + 1$ to N queen at $2(i - N/2) - 1$

1	2
---	---

3	4
---	---

5

$$n = 6$$



$$i = 1 \text{ to } \frac{N}{2} \text{ even col}$$

$$= 1, 2, 3 \dots (n, x) \text{ row}$$

$$i = \frac{N+1}{2} \text{ to } N \text{ at } j = \lfloor x \rfloor$$

$$2(i - \frac{N}{2})$$

Q ₄	Q ₁	Q ₅	Q ₂	Q ₆	Q ₃
----------------	----------------	----------------	----------------	----------------	----------------

$$n = 6$$

$$[r \dots 1] \times 1, 2, 3 \dots$$

--	--	--	--	--	--	--	--

$$(r, k+x) \text{ group } 1, 5, 6, 7$$

$$n = 7$$

Subset sum problem

$$A = \{1, 2, 5, 6, 8\} \quad d = 9$$

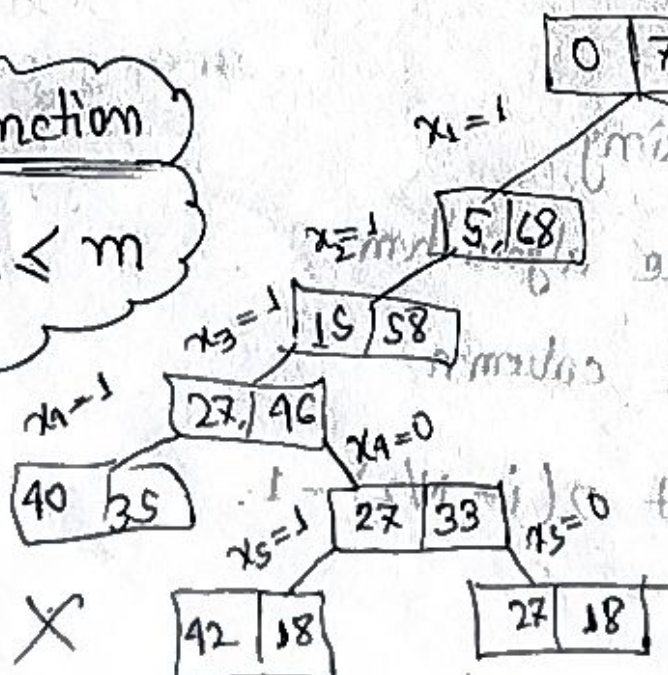
sum will be 9

$$\{1, 2, 6\} \quad \{1, 8\}$$

$$d = \sum_{i=1}^n A_i x_i$$

bounding function

$$\sum_{i=1}^n A_i x_i + A_{k+1} \leq m$$



$x_4 = 0$ first value 27 sum (27) $(1 - 46 - 0) x_4 = 33$

$$A = \{5, 8, 10, 12, 13, 15, 18\}$$

$$n = 6 \quad m = 30$$

$$0 \quad 73.5 \quad \text{sum}$$

$$x_1 = 1$$

$$x_2 = 1$$

$$x_3 = 1$$

$$x_4 = 1$$

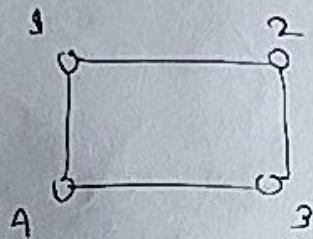
$$x_5 = 0$$

$$x_6 = 1$$

$$x_7 = 0$$

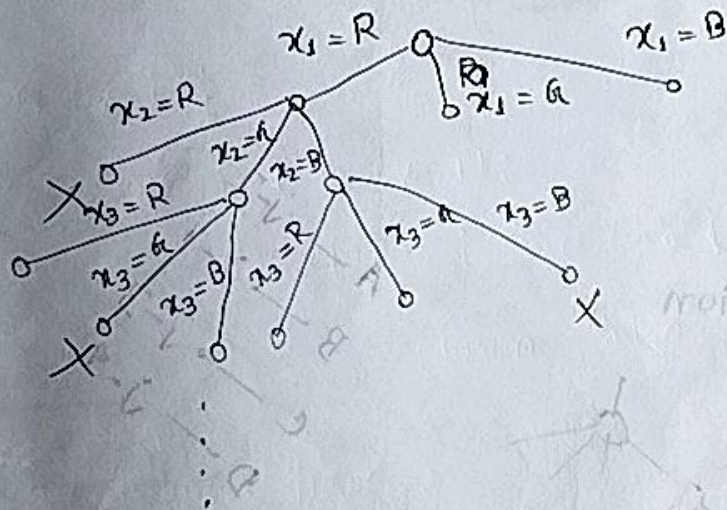
$$\sum_{i=1}^k A_i x_i + \sum_{i=k+1}^n A_i x_i > m$$

n-coloring problem



$\{R, G, B\}$

Adjacent $\times$ same color



Branch and Bound

	1	2	3	4	5
1	0	1	1	1	1
2	1	0	1	1	1
3	1	1	0	1	1
4	1	1	1	0	1
5	1	1	1	1	0

~~db.py~~ →

db.py → mongoDB instance

↓
Singleton pattern

teacher, student, staff

└─ Factory pattern

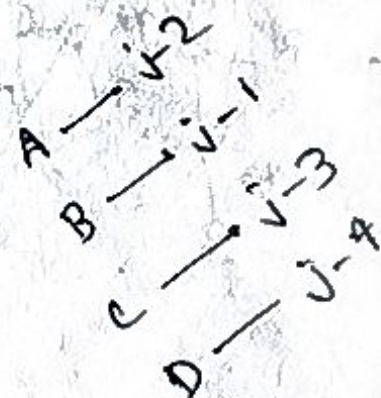
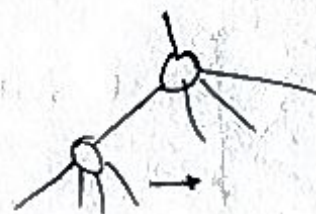
NP → Hamiltonian → Approximation

Branch and Bound

→ Optimization problem
→ Minimization / Maximization

BFS approach

job assigning problem



Person	job-1	j-2	j3	j4
A	9	2	7	8
B	6	4	3	7
C	5	8	1	8
D	7	6	9	4

∴ lb = 10

Assign job-1 to A

9
3
1
4
lb = 17

job-2 → A

job-3 → A

job-4 → A

2			
3			
1			
4	20/10	20/10	20/10
10 = 10	ol	op	1
	2	5	5
	3	2	2
	A	5	A

7			
4			
5			
4			
20 = 20	ol	op	1
	2	5	5
	3	2	2
	A	5	A

8			
3			
1			
45			
18 = 18	ol	op	1
	2	5	5
	3	2	2
	A	5	A

A will take job-2

6	4	X	7
5	8	X	8
7	6	X	4

min in all rows.

ol = v, ol = w
ol = du

for B

B → job-1

$$2 + 6 + 1 + 4 = 13$$

B → job-3

$$2 + 3 + 8 + 4 = 17$$

B → job-4

$$2 + 7 + 1 + 7 = 17$$

A will take job-1

for C

1	8
0	9

$$2 + 6 + 1 + 4 = 13$$

for D

$$2 + 6 + 8 + 9 = 25$$

C will take job-3

D will take job-4

Knapsack Problem

upper bound. $ub = v + (w - w_i)(v_{i+1}/w_{i+1})$

$w = 0$

$v = 0$

$ub = 100$

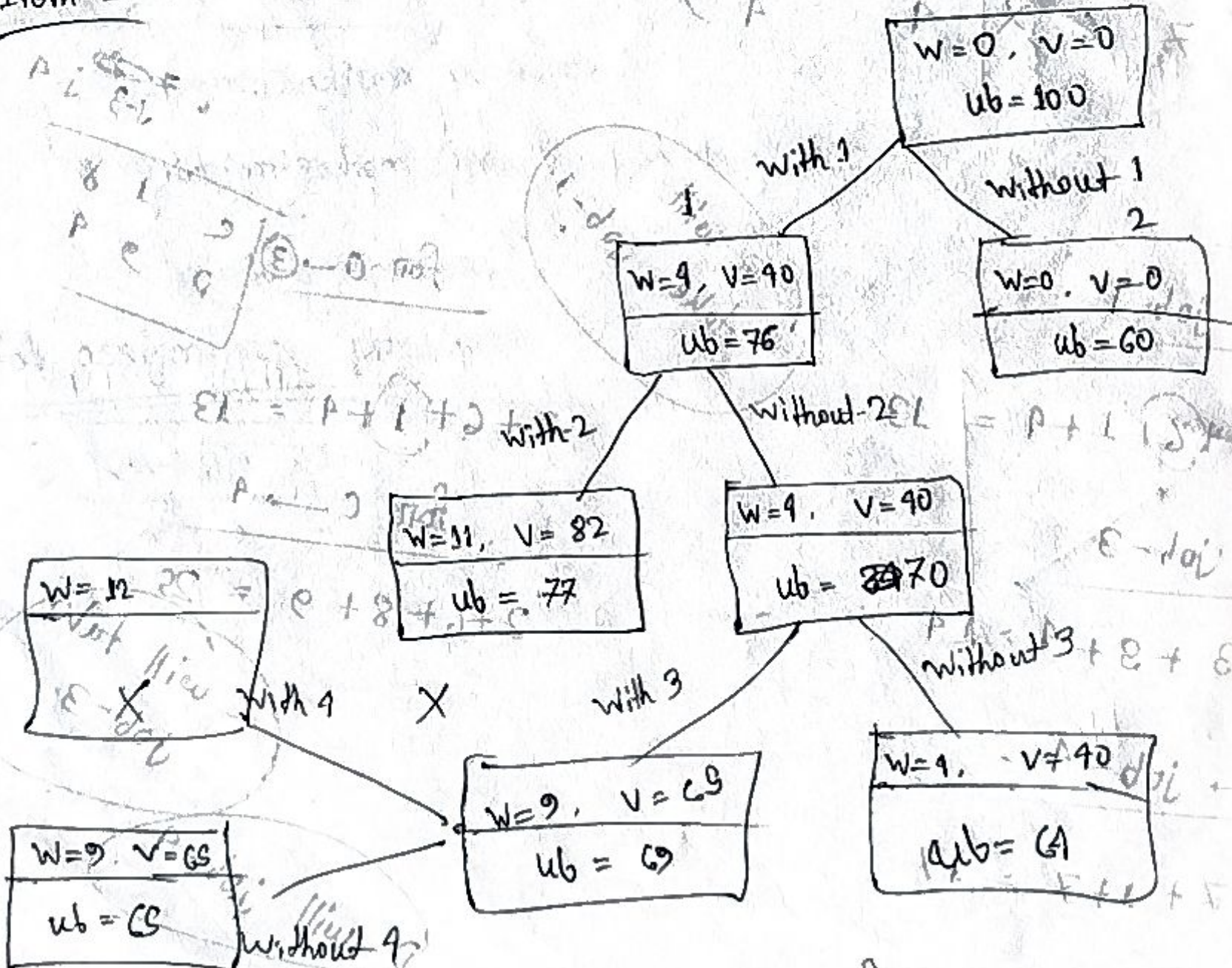
$W = 10$

item	weight	value	value/weight
1	4	40	10
2	7	42	6
3	5	25	5
4	3	12	4

Item-1

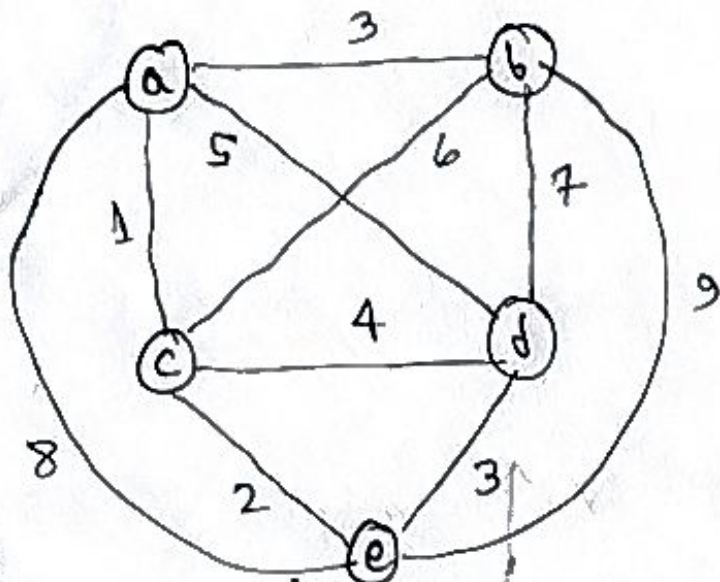
$ub = 40 + (10 - 4)(42/7) = 76$

Item-2



Travelling Sales person problem

Handwritten notes



a =

Handwritten notes

Handwritten notes

$$D = \begin{bmatrix} 0 & 3 & 5 & 6 & 8 \\ 3 & 0 & 4 & 7 & 9 \\ 5 & 4 & 0 & 2 & 3 \\ 6 & 7 & 2 & 0 & 3 \\ 8 & 9 & 3 & 3 & 0 \end{bmatrix}$$

$$(1+3) + (3+6) + (1+2) + (4+3) + (2+3)$$

2

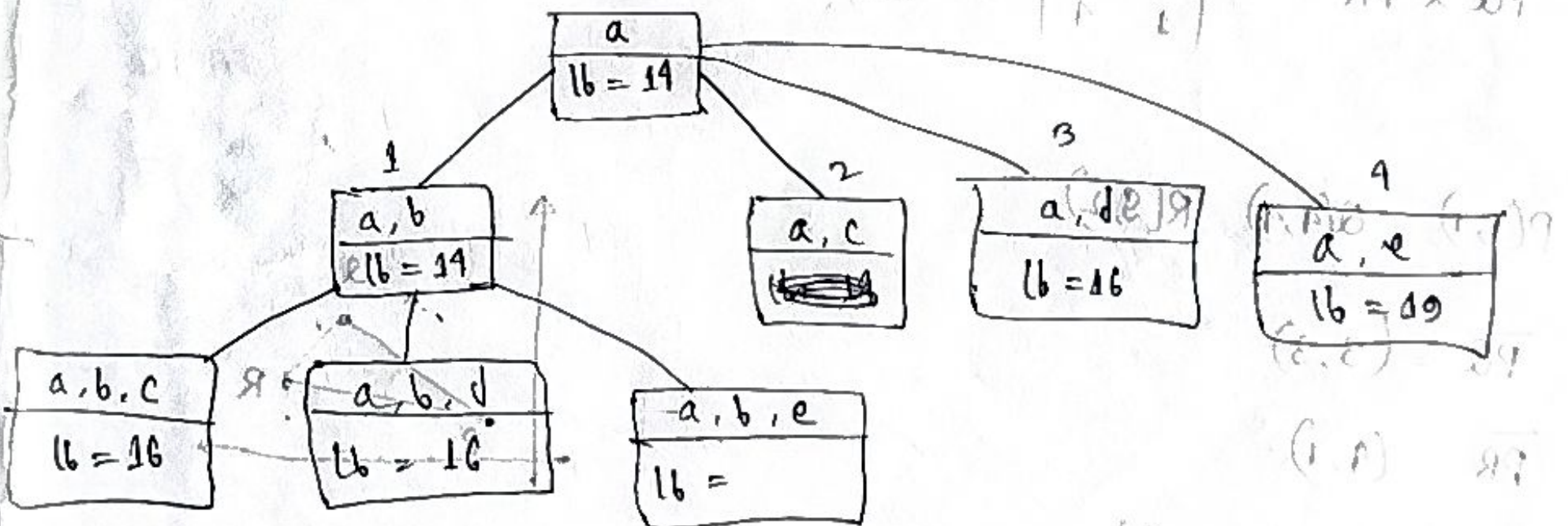
Handwritten notes

$$= \frac{4+9+3+7+5}{2} = 14$$

$$0 < 8 = 1 - 51$$

$$\begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix} = 2 \times 1 - 5 \times 1 = -3$$

$$89 \times 109$$



Handwritten notes

$$e = 1 - 51 = -50$$

Convex half

smallest enclosing circle problem

set of points — input

convex polygon — output

S A O

E O A

• O E S

angles < 180°

P(1,1), Q(4,2), R(5,5)
(E+1) + (E+1) + (S+1) + (2+E) + (E+1)

$$\overline{PQ} = (3,1)$$

$$\overline{PR} = (4,4)$$

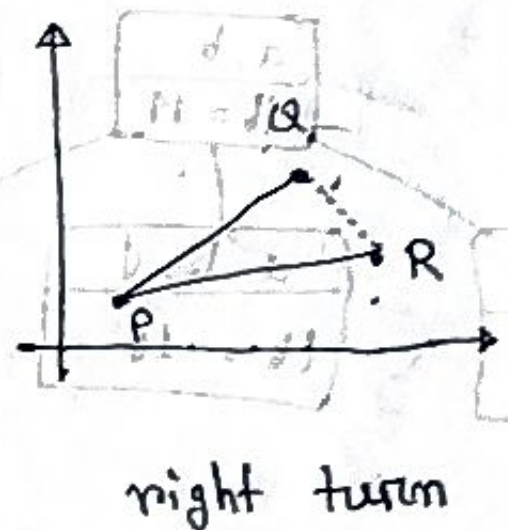
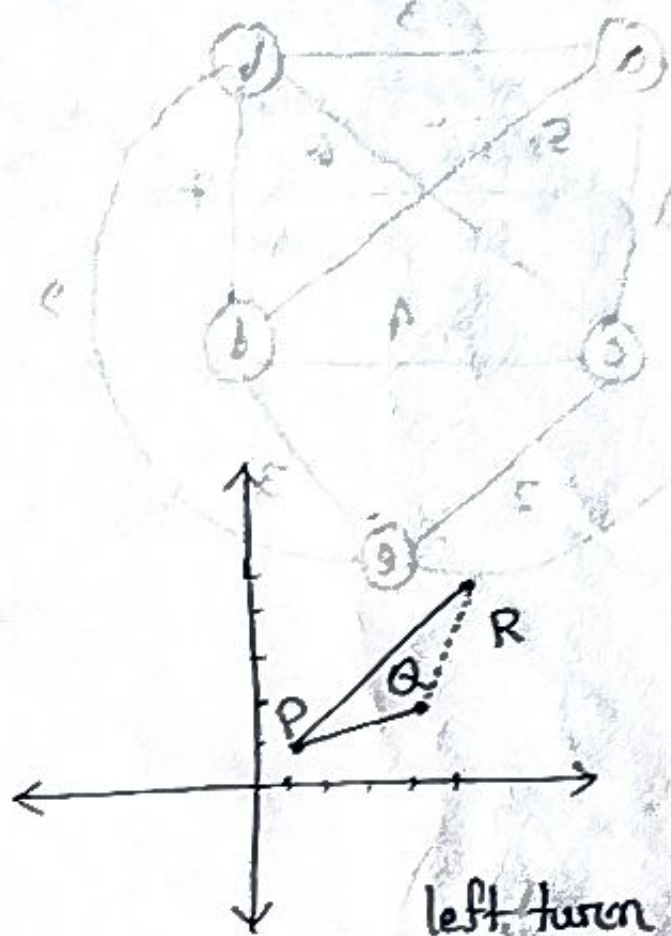
$$\overline{PQ} \times \overline{PR} = \begin{vmatrix} 3 & 4 \\ 1 & 4 \end{vmatrix} = 12 - 4 = 8 > 0$$

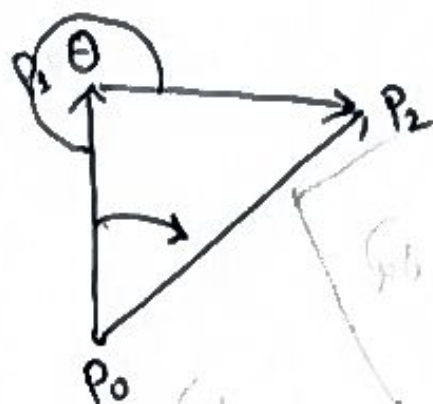
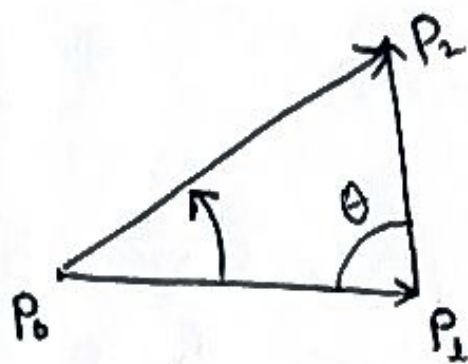
P(1,1), Q(4,4), R(5,2)

$$\overline{PQ} = (3,3)$$

$$\overline{PR} = (4,1)$$

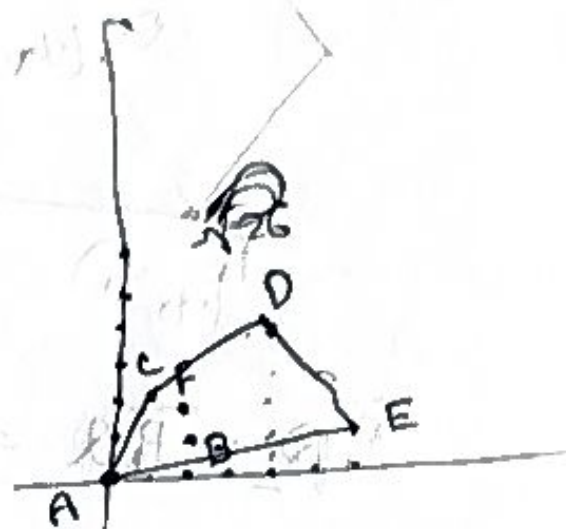
$$\overline{PQ} \times \overline{PR} = \begin{vmatrix} 3 & 4 \\ 3 & 1 \end{vmatrix} = 3 - 12 = -9$$





left turn $\longrightarrow \theta < 180^\circ$

right turn $\longrightarrow \theta > 180^\circ$



$A(0,0)$, $B(2,1)$, $C(1,3)$, $D(4,4)$, $E(5,0)$, $F(2,2)$

$A(0,0)$ $A(0,0^\circ)$

$B(\sqrt{5}, 26.565^\circ)$

$C(\sqrt{10}, 71.565^\circ)$

$D(4\sqrt{2}, 45^\circ)$

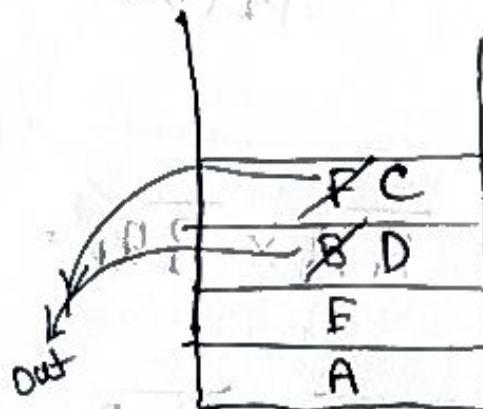
$E(\sqrt{26}, 11.31^\circ)$

$F(2\sqrt{2}, 45^\circ)$

$\longrightarrow A < E < B < D < F < C$

$F(2,2)$

A



$$\begin{vmatrix} -1 & -3 \\ 3 & 1 \end{vmatrix} = -1 + 9 = 8$$

$$\overline{FD} \times \overline{FE}$$

$$\begin{vmatrix} 2 & 3 \\ 2 & 1 \end{vmatrix} = 2 - 6 = -8$$

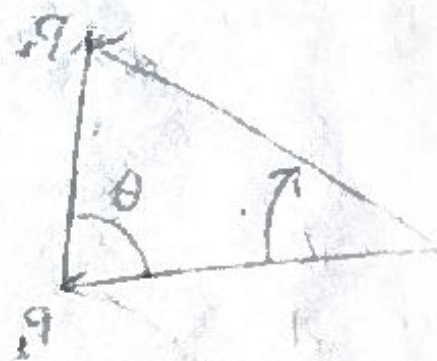
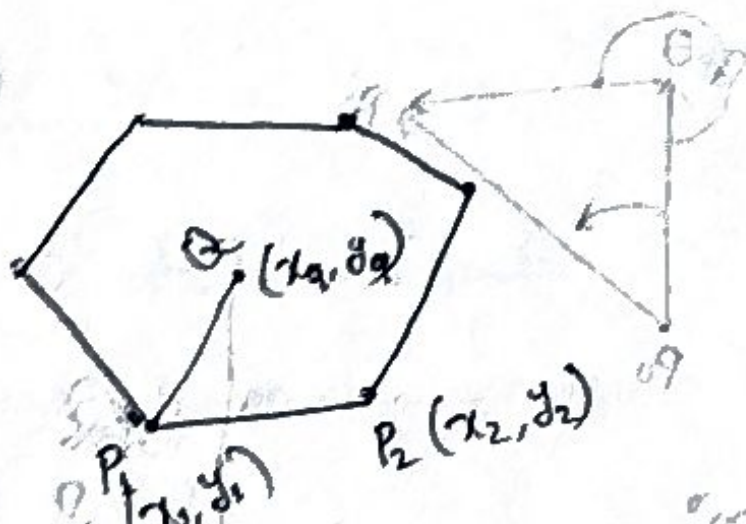
$$\overline{EB} = (-3, 0)$$

$$\overline{ED} = (-1, 3)$$

$$\overline{EB} \times \overline{ED} = \begin{vmatrix} -3 & -1 \\ 0 & 3 \end{vmatrix} = -9$$

right turn

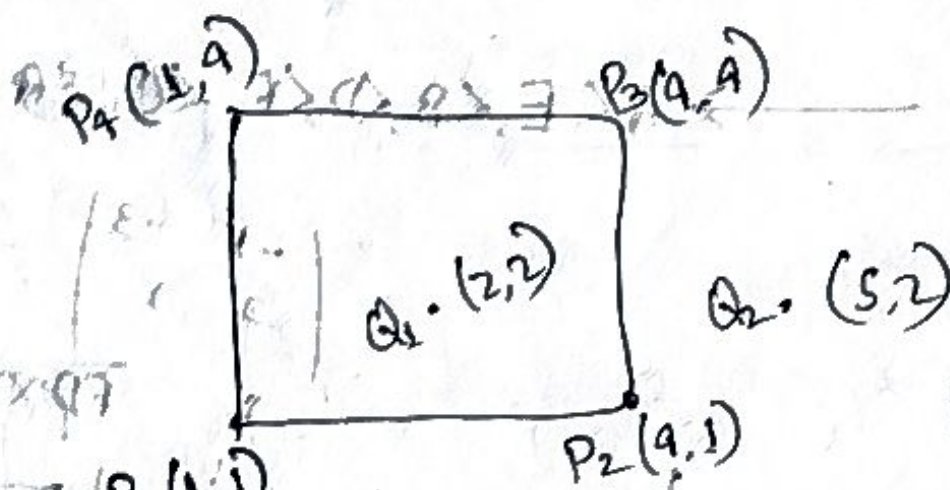
$$\begin{array}{cccc} \overline{ED} & \overline{EF} & \overline{DF} & \overline{DC} \\ \downarrow & \downarrow & \searrow & \searrow \\ (-1, 3) & (-3, 1) & (2, 2) & (-3, 1) \end{array}$$



$$\overrightarrow{P_1 P_2} \times \overrightarrow{P_1 Q} = (x_2 - x_1, y_2 - y_1) \times (x_q - x_1, y_q - y_1)$$

normal vector

if all sign are same or zero, point is inside



$$\overrightarrow{P_1 P_2} \times \overrightarrow{P_1 Q_1}$$

$$\overrightarrow{P_1 P_2} \times \overrightarrow{P_1 Q_1} = \begin{vmatrix} 3 & 1 \\ 0 & 1 \end{vmatrix} = 3$$

$$\overrightarrow{P_2 P_3} \times \overrightarrow{P_2 Q_1} = \begin{vmatrix} 0 & -2 \\ 3 & 1 \end{vmatrix} = 6$$

$$\overrightarrow{P_3 P_4} \times \overrightarrow{P_3 Q_1} = \begin{vmatrix} -3 & -2 \\ 0 & 1 \end{vmatrix} = 6$$

$$\overrightarrow{P_4 P_1} \times \overrightarrow{P_4 Q_1} = \begin{vmatrix} 0 & 1 \\ -3 & -2 \end{vmatrix} = 3$$

all are positive, so Q is inside.

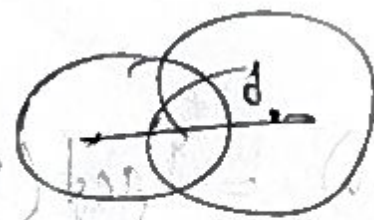
Q_2

$d < r_1 + r_2$

$$\overline{P_1 P_2} \times \overline{P_1 Q_2} = \begin{vmatrix} 3 & 4 \\ 0 & 1 \end{vmatrix} = 3$$

$r_1 > r_2$

$$\overline{P_2 P_3} \times \overline{P_2 Q_2} = \begin{vmatrix} 0 & 1 \\ 3 & 1 \end{vmatrix} = -3$$



$d < r_1 + r_2$



$d < r_2 < r_1$

as the sign are different, so Q_2 outside.

$$C_1 = (x_1, y_1)$$

$$C_1 = (x_1, y_1)$$

$$Q_3(3, 1)$$

$$\overline{P_1 P_2} \times \overline{P_1 Q_3} = \begin{vmatrix} 3 & 2 \\ 0 & 0 \end{vmatrix} = 0$$

in the boundary line.

$$\overline{P_2 P_3} \times \overline{P_2 Q_3} = \begin{vmatrix} 0 & 0 \\ 3 & 1 \end{vmatrix} = 0$$

shoelace formula

$$\text{Area} = \frac{1}{2} \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i)$$

$$\frac{1}{2} \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i)$$

$$= \frac{1}{2} \begin{vmatrix} 1 & 4 & 4 & 1 & 1 \\ 1 & 1 & 4 & 4 & 1 \end{vmatrix} = \frac{1}{2} (1 + 16 + 16 + 1 - 4 - 4 - 4 - 4)$$

9 unit²

$$\text{gcd}(a, b) = \text{gcd}(b, a \bmod b)$$

$$a = c + b \times k$$

$$\therefore a \bmod b = c$$

$$\text{gcd}(b, a \bmod b)$$

$$\text{gcd}(b, c)$$

$$\text{gcd}(a, b)$$

$$= \text{gcd}(c + b \times k, b)$$

$$\text{gcd}(b, b)$$

$$\text{gcd}(c, b)$$

$$\text{gcd}(a, b) = \text{gcd}(b, a \bmod b)$$

$$a \mid b \ \& \ b \mid c \longrightarrow a = \pm b$$

$$\text{gcd}(a, b) \mid \text{gcd}(b, a \bmod b)$$

$$\text{gcd}(b, a \bmod b) \mid \text{gcd}(a, b)$$

$$\text{gcd}(a, b) = d$$

$$d \mid a, d \mid b \longrightarrow d \mid ax + by$$

$$d \mid a \bmod b$$

$$\text{as, } a \bmod b = a - \left\lfloor \frac{a}{b} \right\rfloor b$$

while (r != 0)

$$r = a \% b$$

$$a = b$$

$$b = r$$

return b

$$g\left(\begin{smallmatrix} 52 & 37 \\ 38 & 52 \end{smallmatrix}\right) = 19$$

$$g(38, 19) = 19$$

$$a \bmod b = a - \left\lfloor \frac{a}{b} \right\rfloor b$$

$$d \mid a \ \& \ d \mid b$$

$$d \mid ax + by$$

$$d \mid a \bmod b$$

$$\gcd(b, a \bmod b) \mid \gcd(a, b)$$

$$d \mid b \quad d \mid a \bmod b$$

$$a = qb + (a \bmod b)$$

$$a = \frac{a}{b} + \pm b$$

$$d \mid \gcd(a, b)$$

$$a = 99, \quad b = 78$$

Extended_euclid(a, b)

if b == 0

return(a, 1, 0)

else (d', x', y') = Extended_euclid(b, a mod b)

(d, x, y) = (d', y', x' - [a/b] y')

return(d, x, y)

Extended Euclid Algorithm

$$\gcd(a, b) = d$$

$$d = ax + by$$

$$78 \mid 99 \mid 1$$

$$\begin{array}{r} 2 \mid 78 \mid 3 \\ 63 \end{array}$$

$$\begin{array}{r} 15 \mid 21 \\ 63 \end{array}$$

$$\begin{array}{r} 6 \mid 15 \mid 2 \\ 12 \\ 3 \mid 6 \end{array}$$

$$\gcd = 3$$

Online Algorithms

1. Keep renting
2. Buy on first day
3. Rent on first two day and buy on third day.

$$\text{day-1 } \frac{500}{500} = 1$$

$$\text{d-2 } \frac{1000}{1000} = 1$$

$$\text{d-3 } \frac{1500}{1500} = 1$$

$$\text{d-4 } \frac{2000}{1500} = 1.3333$$

$$\frac{500 \times m}{1500}$$

Buy on first day

$$\text{d-1} = \frac{1500}{500} = 3$$

$$\text{d-2} = \frac{1800}{1000} = 1.8$$

$$\text{d-3} = \frac{1800}{1500} = 1.2$$

Optimal

then all are 1

Rent first 2, then buy

$$\text{d-1} = \frac{500}{500} = 1$$

$$\text{d-2} = \frac{1000}{1000} = 1$$

$$\text{d-3} = \frac{1500}{1500} = 1$$

$$\text{d-4} = \frac{2000}{1500} = 1.33$$

all are 1.33

$$T(i) = k + (k - x_i + y_i - d) - (x_i + y_i)$$

$$= 2k - 2x_i - d$$

$$\leq 2k - 2x_i$$

$$\leq 2i$$

How $(k - x_i) \rightarrow i$?

$$T(m) = k_m + \text{inv}(m+1) - \text{inv}(m)$$

inverse

10 ~~XXXXX~~

- online algo $\rightarrow$ comparative ratio
best " " (Rent-purchase)
math 9/17

$$\frac{mn}{\frac{m(n+1)}{2}}$$

Linear search
algorithm

$\rightarrow$ Number theory $\rightarrow$ division property (gcd, Ext. gcd)
proof

Fermate, euler, CRP theorem
Mod, inverse.

application
WISIT 300%
 $\hat{=}$

RSA correctness proof

$\rightarrow$ Computational Geometry -

Line intersection,
convex halves,
GRAHAMs algo, vector
traverse right-left,
angle,
polygon.

~~shortest~~ formula,
shoelace